2010年真题参考答案

一、选择题

(1)C. (2)A. (3)B. (4)C. (5)A. (6)D. (7)C. (8)A.

二、填空题

(9) -1. (10) $\frac{\pi^2}{4}$. (11) $pe^{\frac{p^3-1}{3}}$. (12) 3. (13) 3. (14) $\sigma^2+\mu^2$.

三、解答题

(15) e^{-1}.

(16) $\frac{14}{15}$.

(17) 最大值为 $5\sqrt{5}$,最小值为 $-5\sqrt{5}$.

(18) (Ⅰ)$\int_0^1 |\ln t|[\ln(1+t)]^n dt < \int_0^1 t^n |\ln t| dt (n=1,2,\cdots)$.

(Ⅱ)$\lim\limits_{n\to\infty} u_n = 0$.

(19) 证明略.

(20) (Ⅰ)$\lambda=-1$, $a=-2$.

(Ⅱ)$k(1,0,1)^{\mathrm{T}} + \left(\frac{3}{2}, -\frac{1}{2}, 0\right)^{\mathrm{T}}$ 为 $\boldsymbol{Ax}=\boldsymbol{b}$ 的通解,其中 k 为任意常数.

(21) $a=-1$, $\boldsymbol{Q}=\begin{pmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ \frac{2}{\sqrt{6}} & 0 & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{pmatrix}$, $\boldsymbol{Q}^{\mathrm{T}}\boldsymbol{AQ}=\begin{pmatrix} 2 & 0 & 0 \\ 0 & -4 & 0 \\ 0 & 0 & 5 \end{pmatrix}$.

(22) $A=\frac{1}{\pi}$, $f_{Y|X}(y|x)=\frac{1}{\sqrt{\pi}}e^{-(x-y)^2}$, $y\in(-\infty,+\infty)$.

(23) (Ⅰ)随机变量(X, Y)的概率分布为

X \ Y	0	1	2
0	$\frac{1}{5}$	$\frac{2}{5}$	$\frac{1}{15}$
1	$\frac{1}{5}$	$\frac{2}{15}$	0

(Ⅱ)$-\frac{4}{45}$.